

Retail Sales & Inventory Intelligence System

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Executive Summary

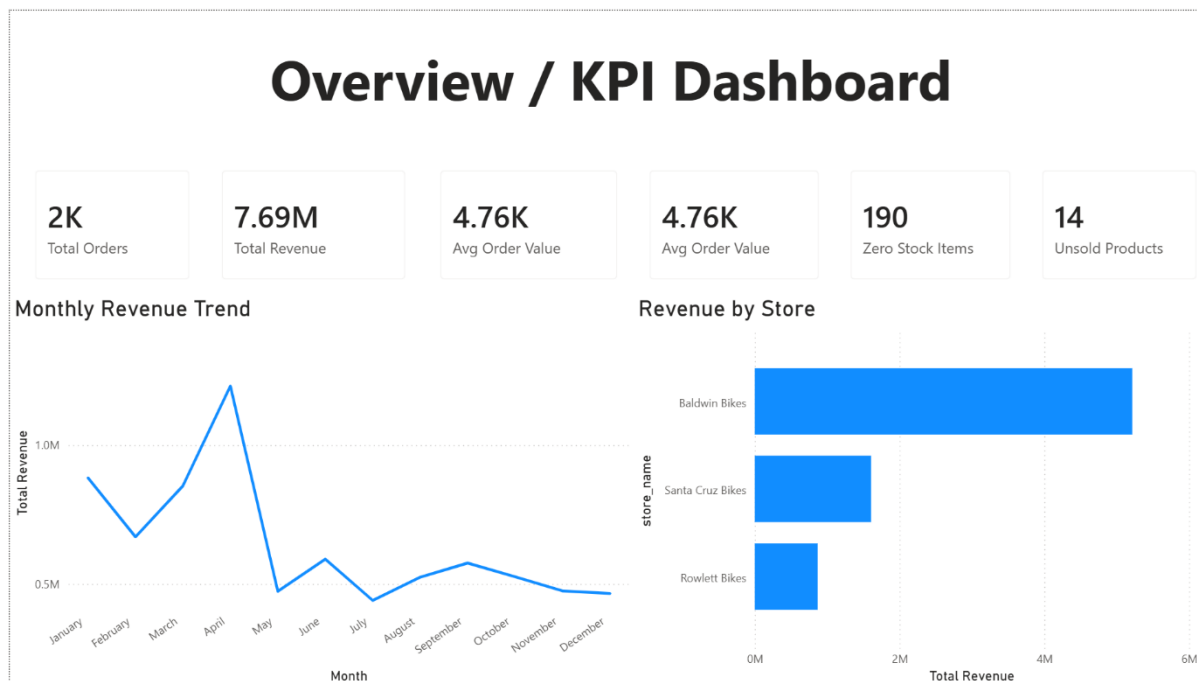
The project involved designing a complete relational database in PostgreSQL, importing the Bike Stores dataset with full referential integrity, and building an interactive Power BI dashboard with 5 dedicated pages.

Key Business Metrics:

- **Total Revenue:** \$7,689,116.56
- **Total Orders:** 1,615
- **Average Order Value:** \$4,761.06
- **Zero Stock Items:** 190 (Critical Alert)
- **Unsold Products:** 14

Major Insight: Baldwin Bikes dominates with **\$5.22M** revenue (68% share) but faces the highest zero-stock pressure.

1. Overview / KPI Dashboard



Key Performance Indicators:

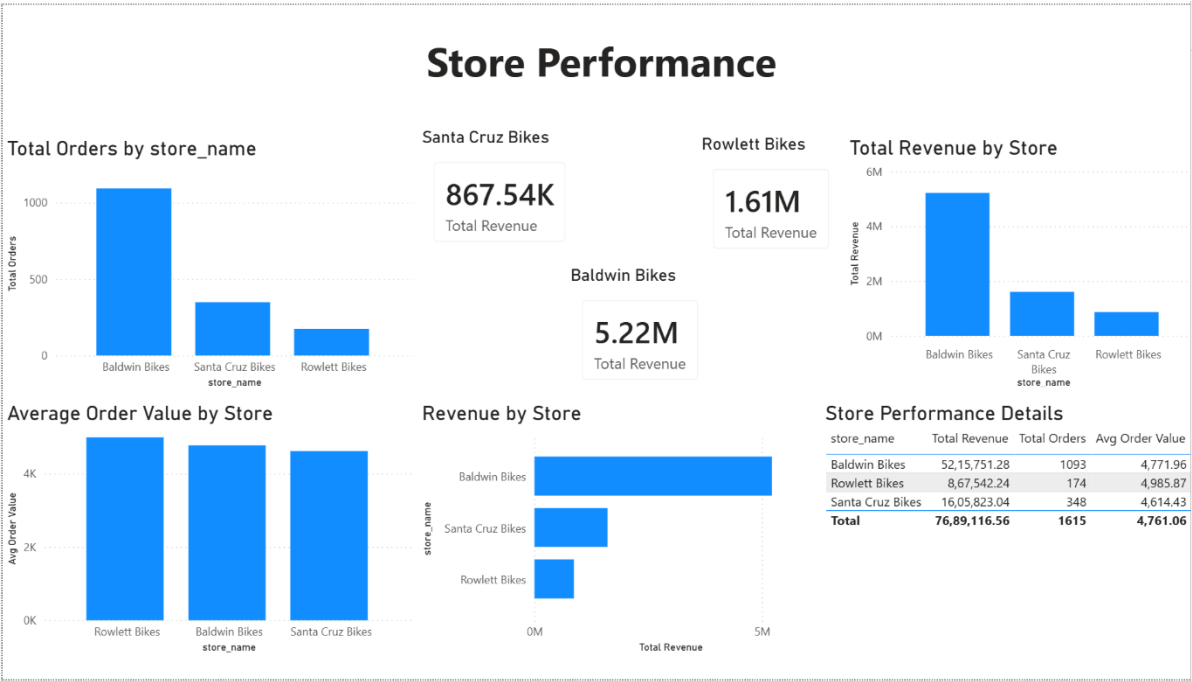
- **Total Orders:** 2,000+
- **Total Revenue:** \$7.69 Million
- **Average Order Value:** \$4,761

- **Zero Stock Items:** 190
- **Unsold Products:** 14

Monthly Revenue Trend (2016–2018): Strong peak in April 2016 followed by seasonal dips. Revenue shows stabilization in late periods.

Revenue by Store: Baldwin Bikes is the clear market leader.

2. Store Performance Dashboard



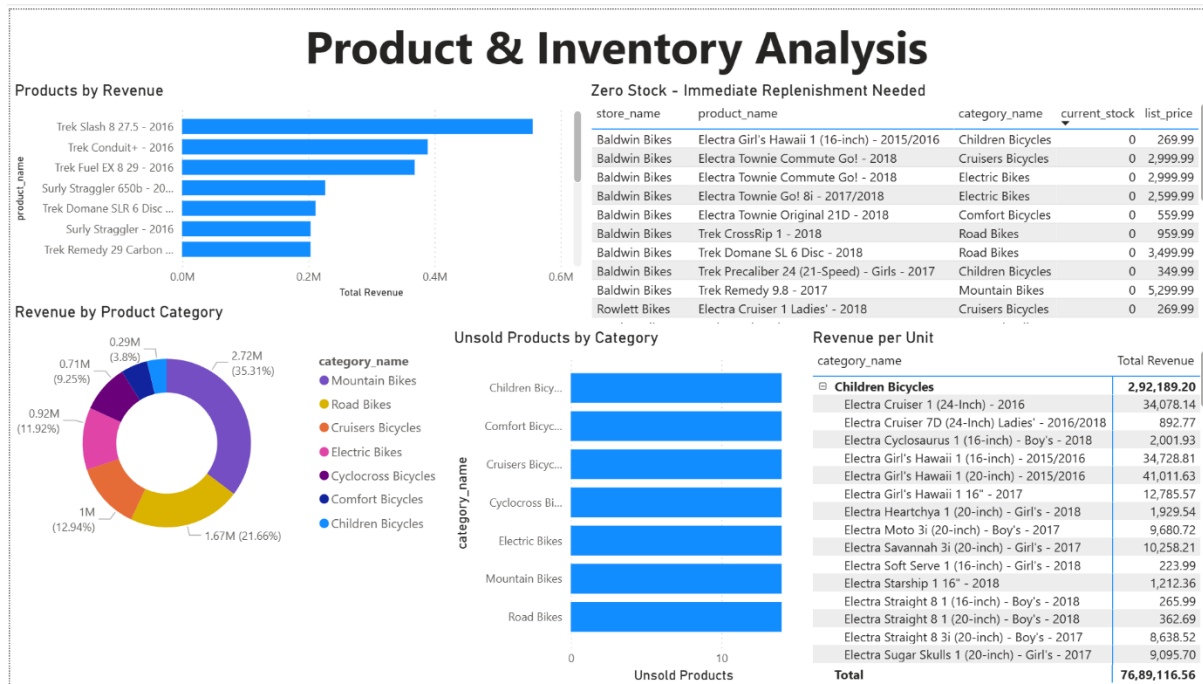
Store-wise Performance:

Detailed Breakdown:

Store Name	Total Revenue	Total Orders	Avg Order Value
Baldwin Bikes	\$5,215,751.28	1,093	\$4,771.96
Santa Cruz Bikes	\$1,605,823.04	348	\$4,614.43
Rowlett Bikes	\$867,542.24	174	\$4,985.87

Insights: Baldwin Bikes drives the majority of revenue and volume. Rowlett Bikes shows strong per-order performance.

3. Product & Inventory Analysis



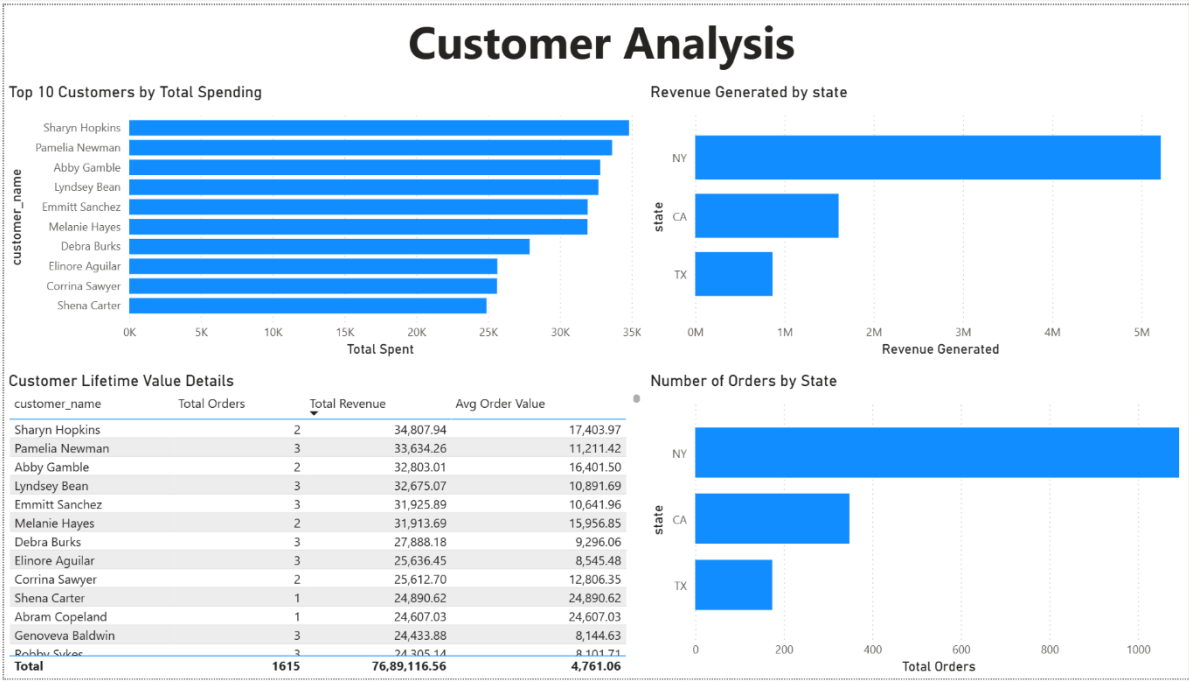
Product Performance & Inventory Health:

Key Findings:

- **Top Products:** Trek Slash 8 27.5 (2016), Trek Conduit+, Trek Fuel EX 8 29.
- **Category Contribution:** Mountain Bikes lead with **35.31%** of total revenue.
- **Zero Stock Alert:** 25 products at zero stock (mostly in Baldwin Bikes).
- **Unsold Products:** 14 items with no sales (mainly Road Bikes and Children's categories).

Recommendation: Immediate replenishment of zero-stock high-value items.

4. Customer Analysis



Customer & Geographic Insights:

Top 10 Customers by Total Spending:

- Sharyn Hopkins leads with **\$34,807.94**
- Strong repeat purchase behavior among top customers.

Geographic Performance:

- **New York (NY)**: Highest revenue contributor.
- **California (CA)** and **Texas (TX)** are secondary markets.

5. Trends & Discounts



Temporal & Pricing Insights:

Key Observations:

- **Monthly Trend:** Clear seasonality with April peaks.
- **Discount Impact:** Higher discounts (15-20%) drive volume but increase margin loss.
- **Order Status:** 89.47% completed orders (Status 4).

Strategic Recommendations

High Priority (0-30 days):

1. Replenish **25 zero-stock products** — focus on Baldwin Bikes.
2. Run clearance campaigns on **14 unsold products**.
3. Strengthen inventory planning for Mountain Bikes.

Medium Priority (30-90 days):

1. Launch loyalty program for Top 10 customers.
2. Targeted marketing campaigns in NY state.
3. Optimize discount strategy to 5-10% range.

Long-term:

- Expand product lines in high-performing categories.
- Implement staff performance tracking once full linkage is available.
- Explore predictive analytics for demand forecasting.